

HACKDEVENGERS 2.0 // ACCESSIBLE ASSISTIVE AI

VoiceChessmate

Eyes-Free, Voice-First Chess Companion for 285M Visually Impaired Players

Stockfish 16 Engine Core • Dual-Path Audio Architecture • FIDE & IBCA Compliance

100% Screen-Free

Full conversational gameplay without a display

< 40ms De-Collision

Zero dropped speech synthesis moves

IBCA Compliant

Official international blind chess phonetics

The Problem: Digital Chess Excludes Blind Players

1. Heavy Visual Lock-In

SYSTEMIC EXCLUSION

- Chess.com & Lichess require continuous 2D visual eye-tracking across 64 squares.
- Board state, clocks, and attack vectors require 100% sight.
- Under Blitz & Rapid clocks, playing without sight leads to immediate flag-fall.
- 285M visually impaired people are shut out from digital competitive play.

2. Screen Reader Friction

SLUGGISH LATENCY

- Generic screen readers (NVDA, VoiceOver) lack chess-specific intelligence.
- They read entire HTML DOM tables cell-by-cell instead of tactical game state.
- Causes 6 to 10 seconds of cognitive delay per move.
- Zero spatial awareness of pins, forks, threats, or open lines.

3. Browser Audio Drops

FATAL RELIABILITY FLAW

- Chromium V8 garbage collection silently drops speech synthesis wrappers mid-sentence.
- Calling cancel() before speak() drops moves due to async Chrome IPC race conditions.
- Missing a single 'Check!' announcement results in immediate game forfeiture.

The Solution: Voice-First Accessible Chess Companion

Conversational Voice Gameplay

NATURAL AUDITORY INTERACTION

- Speak moves naturally: 'Knight to f3', 'Castle kingside', or 'Cesar 4'.
- Instant auditory confirmation with standardized IBCA phonetic coordinates.
- Guaranteed opponent move announcements: 'Opponent responds: Queen David 5. Your turn.'
- Push-to-Talk (Spacebar): Instantly cuts speech output and opens mic without audio collisions.
- Zero visual display required: Complete autonomy from move one to checkmate.

Tactile Spatial Queries & Hotkeys

REAL-TIME SPATIAL INTELLIGENCE

- [D] Describe Board: Full board piece audit or rank-by-rank auditory scan.
- [T] Tactical Threats: Immediate audio breakdown of targeted pieces and hostile forks.
- [O] Own Pieces: Fast summary of player piece locations and active mobility.
- [C] Captured Differential: Instant material balance count (e.g., 'White up +3 pawns').
- Enables rapid, eyes-free mental visualization under timed competitive conditions.

Engineering: 4 Pillars of the Speech Output Guarantee

1. V8 Utterance Garbage Retention

ELIMINATES SPEECH CUTOFF

- Module-level Set<SpeechSynthesisUtterance> retains wrapper objects.
- Prevents Chrome V8 GC from silently cutting off speech mid-sentence.

2. Chrome IPC Cancel-Speak Race Fix

40MS SETTling BUFFER

- Enforces a 40ms asynchronous IPC settling delay between cancel() and speak().
- Guarantees rapid move announcements are never swallowed by browser IPC races.

3. Dual-Path Fallback Coordinator

1200MS AUTOMATIC FALLBACK

- Monitors streaming WebSocket audio in real time.
- Automatically triggers local Web Speech API TTS if stream is silent > 1200ms.

4. Watchdog & 4-Tier Priority Queue

QUEUE PREEMPTION & AUTO-RECOVERY

- Non-leaking watchdog monitors paused state and unfreezes stuck synthesis queues.
- 4-tier queue (Urgent > High > Normal > Low) prioritizes moves over descriptions.

Accessibility Standards: Official FIDE & IBCA Compliance

Official IBCA Phonetic Matrix

FIDE BLIND CHESS STANDARD 2.1

- International Blind Chess Association (IBCA) standard naming:
- Anna (A), Bella (B), Cesar (C), David (D)
- Eva (E), Felix (F), Gustav (G), Hector (H)
- Eliminates acoustic confusion between B/C/D/E/G in loud environments.
- Example: 'b1-c3' is spoken as 'Knight Bella 1 to Cesar 3'.
- Universal standard accepted in blind tournaments globally.

Strict Auditory Narration Contract

UNAMBIGUOUS AUDIO CONTRACT

- Invariant Opponent Move Format: Always prefixes 'Opponent responds: <move>. Your turn.'
- Terminal State Turn Suppression: The cue 'Your turn.' is strictly omitted on Checkmate, Stalemate, or Draw.
- High-Priority Alerts: Immediate verbal priority for Check, Checkmate, and En Passant captures.
- Zero Fluff, Zero Latency: Concise, unambiguous military-style chess communication.

System Architecture: End-to-End Control Flow

1. Sensory Input

INPUT LAYER

- Push-to-Talk microphone stream.
- Tactile single-key hotkeys (D, T, O, C).
- Instant audio cutoff interrupt hook.
- Zero visual UI required.

2. Voice AI

CONVERSATIONAL LAYER

- Low-latency Speech-to-Text parsing.
- Automated Voice Activity Detection.
- Tool calling schema for legal moves.
- Contextual board queries.

3. Chess Core

VALIDATION LAYER

- Stockfish 16 WASM evaluation engine.
- Chess.js legal move validation.
- IBCA phonetic coordinate translator.
- FEN game state manager.

4. Audio Dispatch

GUARANTEED OUTPUT

- Dual-Path Fallback Coordinator.
- V8 GC utterance retention set.
- 40ms IPC settling de-collision.
- Watchdog queue recovery.

Rigorous Quality Assurance: 100% Automated CI Harness

Headless Web Speech Mock Infrastructure

CI/CD TEST HARNESS

- Web Speech API only exists in graphical desktop browsers.
- Built custom MockSpeechSynthesis and MockSpeechSynthesisUtterance for Node.js.
- Accurately reproduces asynchronous audio event loops (start, end, error, voiceschanged).
- Simulates adversarial conditions: rapid cancel-speak collisions and memory garbage collection stress.
- Runs continuously in GitHub Actions CI without needing a display server.

Vitest Automated Pipeline Verification

100% VERIFIED PASS RATE

- Dedicated test suite: speech-and-audio-pipeline.test.ts
- R1 Verified: Utterances survive forced GC without dropping speech handlers.
- R2 Verified: Cancel-speak calls with <40ms delay are de-collided cleanly.
- R3 Verified: Priority queue preemption and PTT interruption audio flush.
- R4 Verified: Dual-path fallback engages within 1200ms timeout.
- R5 Verified: IBCA prefixing verified and turn cue suppressed on game over.
- Result: 221+ Tests Passing (100%) • 0 Errors • 0 Warnings

Roadmap & Vision: Making Chess Truly Accessible Globally

Phase 1: Hackathon Ready

HACKDEVENGERS 2.0 (COMPLETE)

- Completed voice-first offline/online playable engine.
- Stockfish 16 WASM AI opponent with adjustable ELO.
- Dual-path zero-drop audio output guarantee.
- Tactile hotkeys (D, T, O, C) and IBCA phonetic compliance.
- 100% automated test pass in CI.

Phase 2: Global Matches

Q4 2026 MILESTONE

- Live real-time voice matchmaking for visually impaired players.
- Low-latency voice chess clock with auditory increment countdowns.
- Lichess & Chess.com API sync: Play rated games using VoiceChessmate as an accessible client.

Phase 3: Hardware Integration

2027 GLOBAL VISION

- Bluetooth physical electronic chessboard support (DGT, Millennium).
- Haptic sensory edge feedback for illegal moves and check alerts.
- Multilingual phonetic speech engines (Spanish, French, German, Hindi).